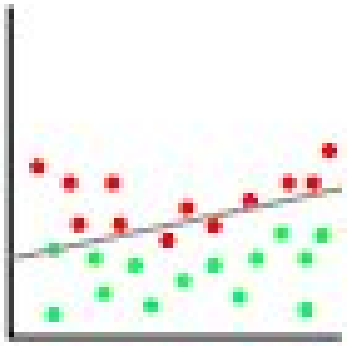
ANN Generalization

Under fitting: A model that fails to sufficiently learn the problem and performs poorly on a training dataset and does not perform well on a holdout sample.

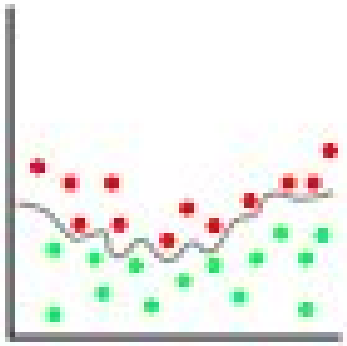
Overfitting: A model that learns the training dataset too well, performing well on the training dataset but does not perform well on a hold out sample.

Good Fit: A model that suitably learns the training dataset and generalizes well to the hold out dataset.

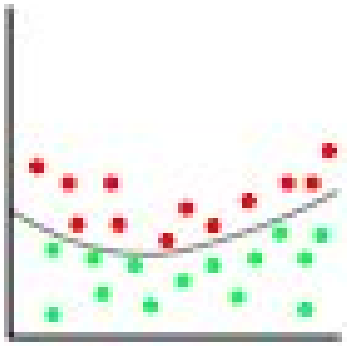
learning & regularization



Underfitting



Overfitting



Balanced

Overfitting - Remedies

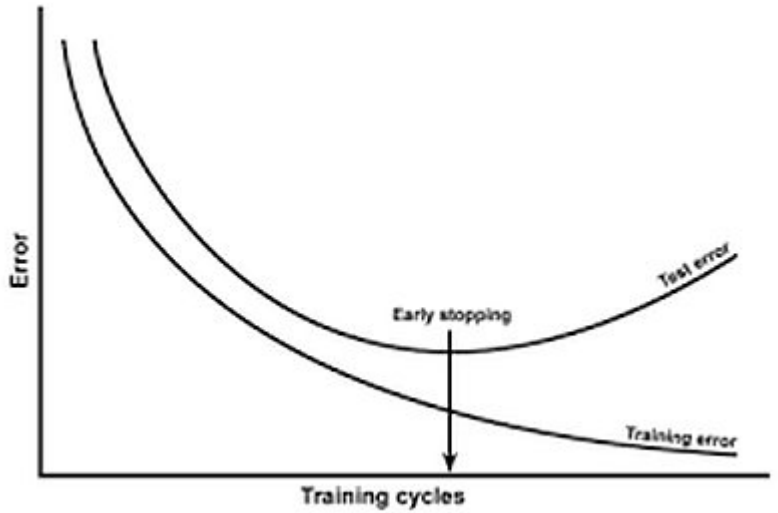
Remedy 1 – Simplifying the Model

Remove layers or reduce the number of neurons to make the network smaller

Overfitting - Remedies

Remedy 2 – Early Stopping

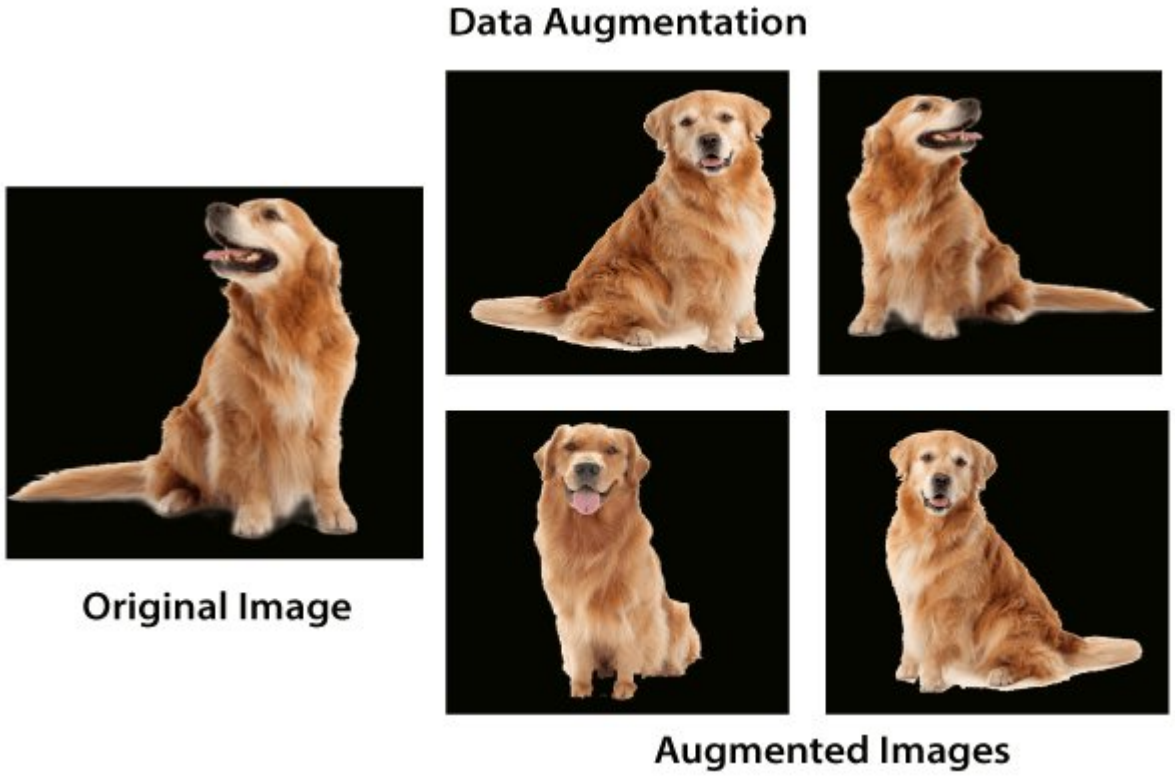
- Form of regularization
- Improves the model's performance on data on the test set
- guidance as to how many iterations can be run before the model begins to overfit



Overfitting - Remedies

Remedy 3 - Use Data Augmentation

Increasing the size of data



Overfitting - Remedies

Remedy 4 – Use Regularization

- Technique to reduce the complexity of the model
- Add a penalty term to the loss function

Overfitting - Remedies

Remedy 4 – Use Regularization

$$L(x,y) \equiv \sum_{i=1}^{n} (y_i - h_\theta(x_i))^2 + \lambda \sum_{i=1}^{n} |\theta_i|$$

L1 Regularization

$$L(x,y) \equiv \sum_{i=1}^{n} (y_i - h_\theta(x_i))^2 + \lambda \sum_{i=1}^{n} \theta_i^2$$

L2 Regularization

Overfitting - Remedies

Remedy 4 – Use Regularization

L1 Regularization	L2 Regularization
1. L1 penalizes sum of absolute values of weights.	1. L2 penalizes sum of square values of weights.
2. L1 generates model that is simple and interpretable.	2. L2 regularization is able to learn complex data patterns.
3. L1 is robust to outliers.	3. L2 is not robust to outliers.

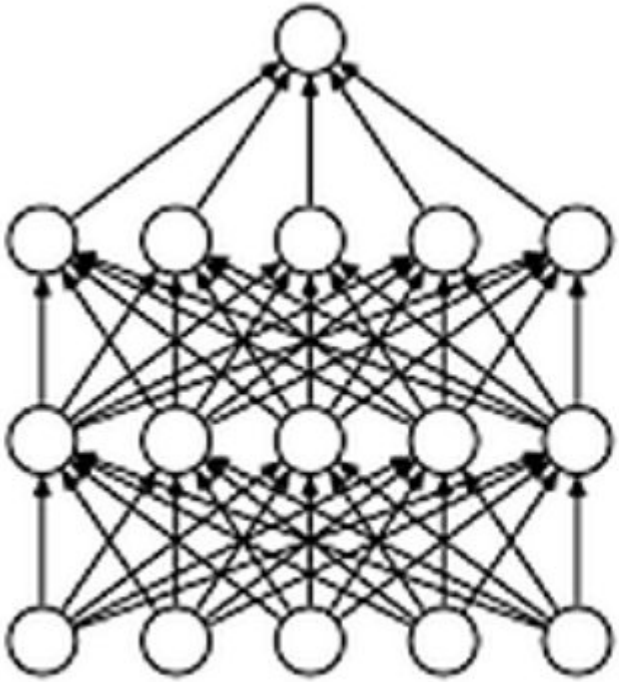
Overfitting - Remedies

Remedy 5 – Dropouts

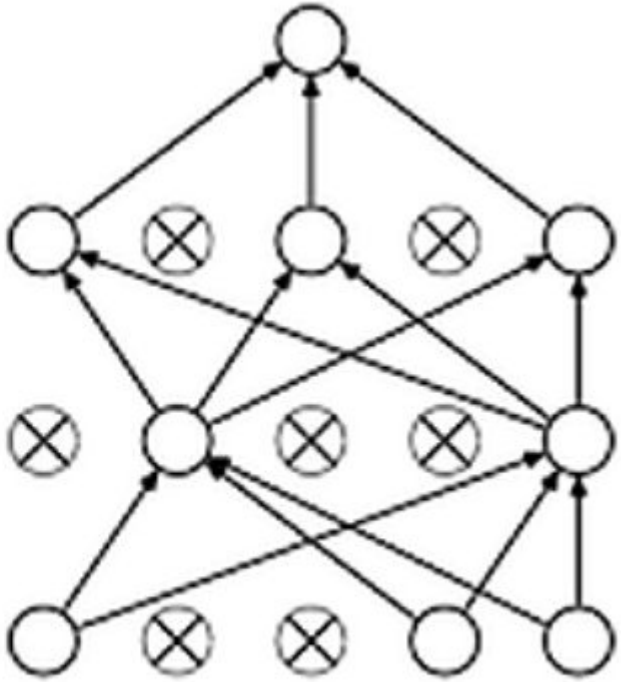
- Modify the ANN
- Randomly drops neurons from the neural network during training in each iteration
- Different networks will over fit in different ways

Overfitting - Remedies

Remedy 5 – Dropouts



(a) Standard Neural Net



(b) After applying dropout.

Overfitting – Easy Quiz

Question 1:

All of the following can help reduce overfitting EXCEPT _____.

i. greater availability of labels
ii. increased data diversity
iii. reduced model complexity
iv. data augmentation

Overfitting – Easy Quiz

Question 2:

In neural network programming, overfitting occurs when a model becomes really good at being able to classify or predict on data that is included in the training set but is not as good at classifying data that it wasn't trained on.

True/ False

Overfitting – Easy Quiz

Question 3:

In neural network programming, if the validation metrics are considerably worse than the training metrics, then this is indication that our model is _____.

i. underfitting
ii. overfitting
iii. generalizing well
iv. diverging with respect to the training set

Overfitting – Easy Quiz

Question 4:

In the concept of neural networks, the concept of overfitting boils down to the fact that the model is unable to _____.

i. learn well
ii. generalize well
iii. predict well
iv. update its weights

Overfitting – Easy Quiz

Question 5:

When training a neural network, we can reduce overfitting by _____.

i. making small changes to the learning rate
ii. adding additional epochs during training
iii. increasing the number of batches for each run
iv. adding data to the training set

Overfitting – Easy Quiz

Question 6:

The general idea of data augmentation involves adding more data to the training set that is similar to the data that we already have but is just reasonably modified to some degree so that it's not the exact same.

True/ False

Overfitting – Easy Quiz

Question 7:

One of the best techniques for reducing overfitting of a neural network is to increase the complexity of the model so that the model will be more capable of extracting patterns in the data.

True/ False

Overfitting – Easy Quiz

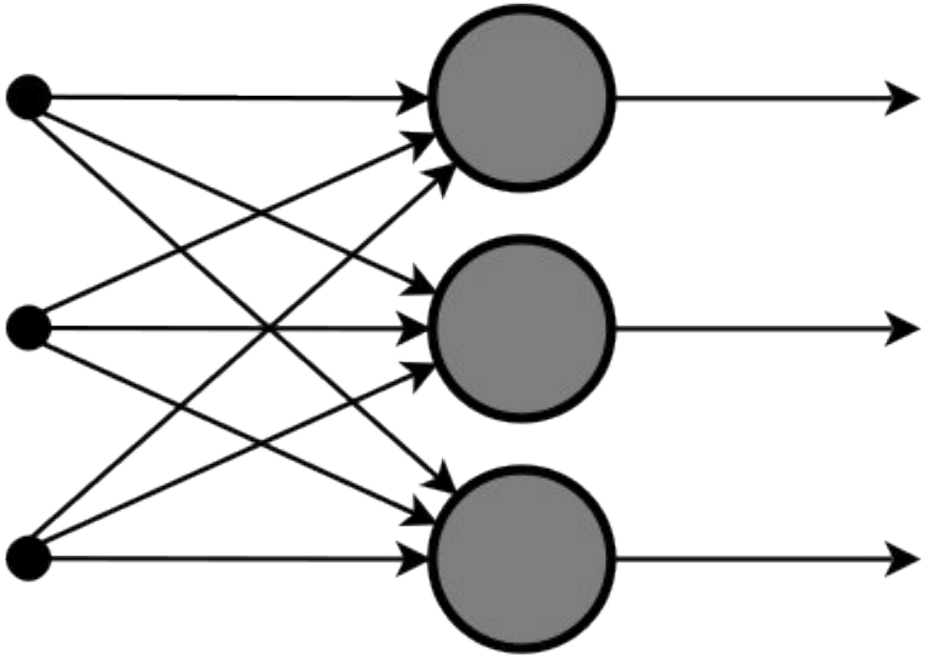
Question 8:

One way of reducing the complexity of a neural network is to remove a layer from the network.

True/ False

TYPES OF NEURAL NETWORK

Feedforward Neural Network (Simplest)



- Data or the input travels in one direction
- Has a front propagated wave and no backpropagation by using a classifying activation function

Feedforward Neural Network (Simplest)

- Applications: Computer vision and speech recognition
- Responsive to noisy data
- Easy to maintain

Radial Basis Function (RBF) Neural Network

- **RBF** is a real valued function whose value depends only on the **distance** between the input and some fixed point, either
  - the origin or
  - some other fixed point c, called **center**

- **Distance** - **Euclidean distance** (some other metrics are used depending on the situation)

- **Applications:**

bathymetry (ocean depth measurement), topography (altitude measurements), hydrology (rainfall interpolation)

RBF ANN



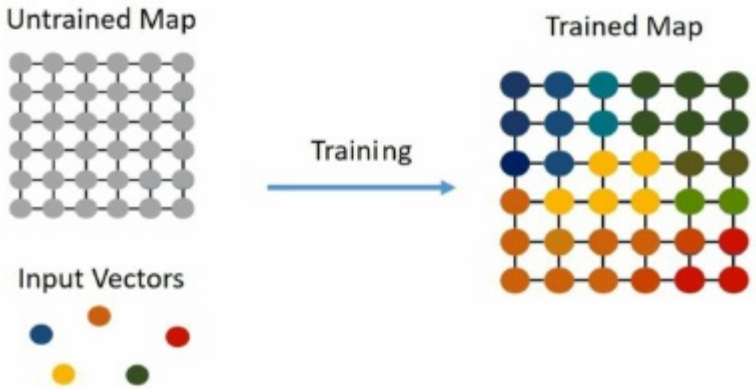
RBF - Applications

- Bathymetry (ocean depth measurement)
- Topography (altitude measurements)
- Hydrology (rainfall interpolation)
- Image warping (digitally manipulating an image)
- Medical Imaging (create images of various parts of the human body)
- etc…

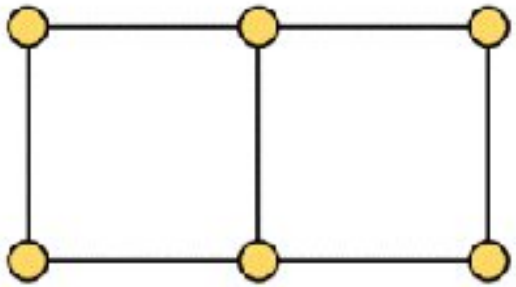
Kohonen Self Organizing Map

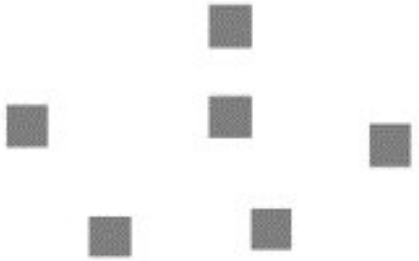
- First, every neuron value is initialized with a small weight and the input vector.

- The neuron closest to the point is the 'winning neuron' and the neurons connected to the winning neuron will also move towards the point



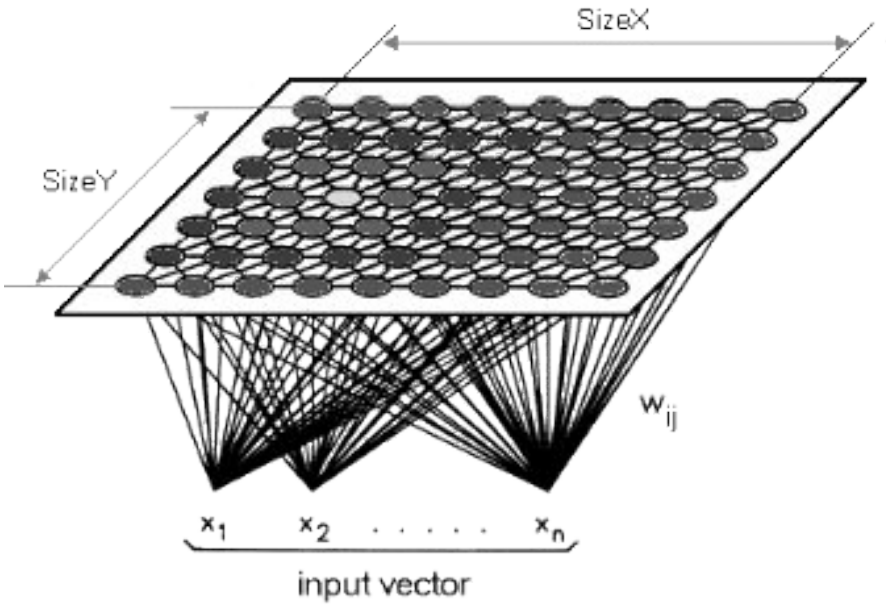
Kohonen SOM

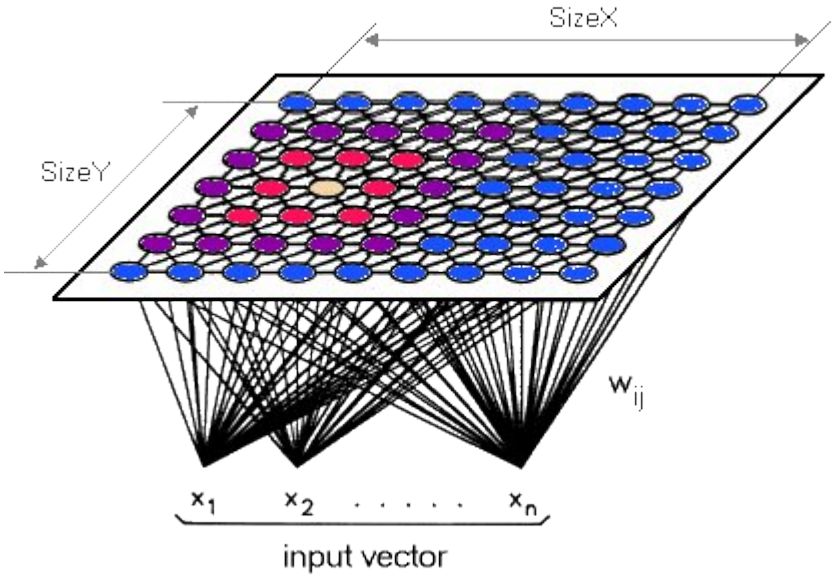




Step 0: Position neurons (orange) in data space.

Kohonen SOM


SizeX
SizeY
w_{ij}
x_1 x_2 . . . . . x_n
input vector


SizeX
SizeY
w_{ij}
x_1 x_2 . . . . . x_n
input vector

Kohonen SOM - Applications

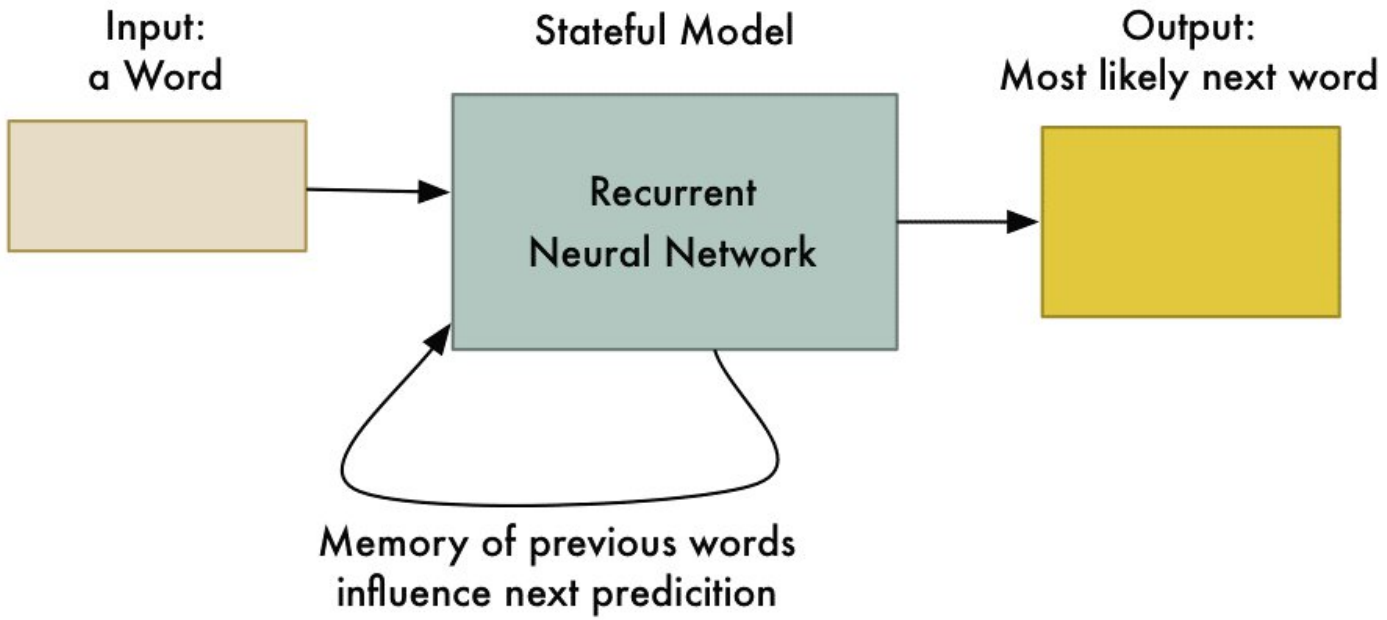
- Clustering data in to different categories
- Medical analysis
- Weather analysis
- etc…

Recurrent Neural Network(RNN) – Long Short Term Memory (LSTM)

Save the output of a layer and feed it back to the input to help in predicting the outcome of the layer.

- First layer – similar to feedforward
- From one time step to the next each neuron will remember some information it had in the previous time-step.
- Each neuron act like a memory cell in performing computations.

Recurrent Neural Network(RNN) – Long Short Term Memory (LSTM)



Output so far:

Machine

RNN - Applications

- Time series prediction.
- Speech recognition.
- Speech synthesis.
- Time series anomaly detection.
- etc…

Modular Neural Network

- Collection of different networks working independently and contributing towards the output.

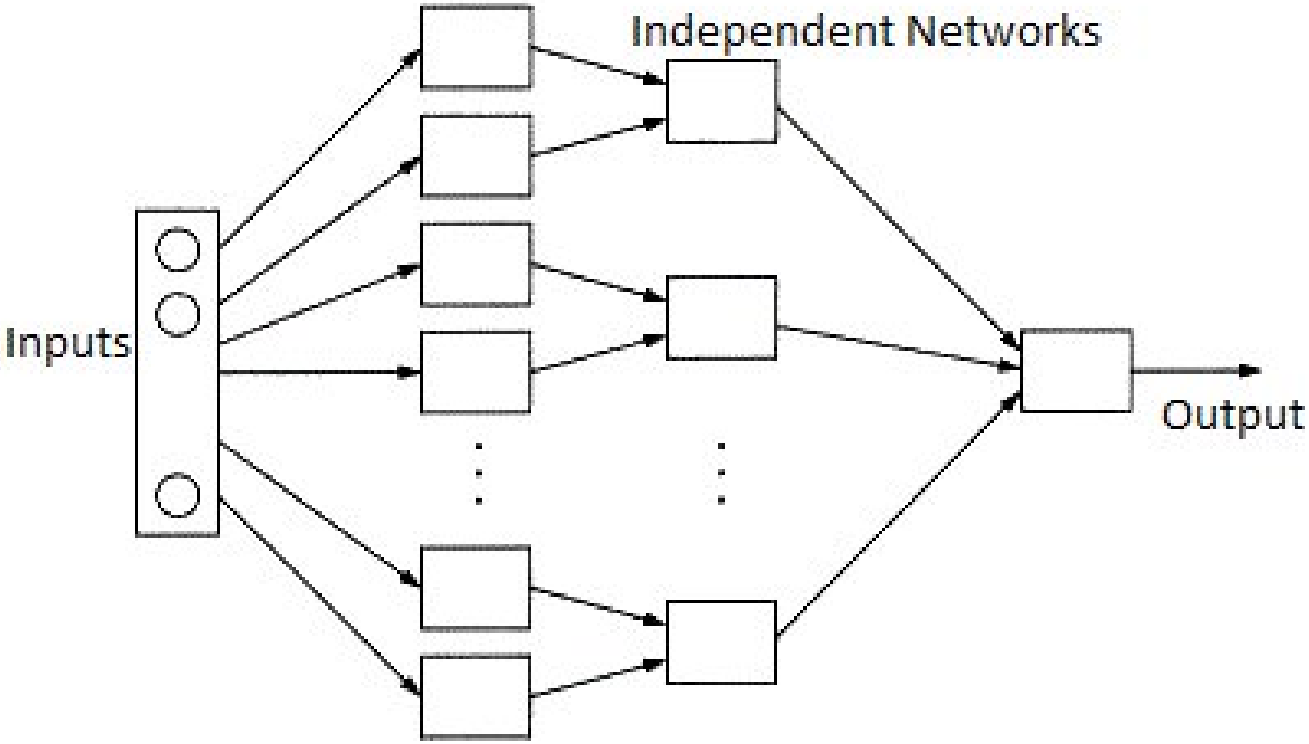
- Each neural network has a set of inputs that are unique compared to other networks constructing and performing sub-tasks.

Modular Neural Network

ADVANTAGE:

- Breakdowns a large computational process into smaller components decreasing the complexity
- High computational speed

Modular Neural Network



ANN in Practice

- Open source libraries used
  - Keras
  - Tensorflow
  - PyTorch
  - etc…

- **Keras -** one of the well-known APIs that is open-source, with a neural network library written in Python.

- **Tensorflow -** open-source Machine Learning library meant for analytical computing. It is a cross-platform tool.

Keras vs. Tensorflow

Parameters	Keras	TensorFlow
Origin	A simple Python library for Deep Learning, developed by Francois Chollet	A set of libraries for Machine Learning, developed by Google Brain
Architecture	Concise and readable with pure architecture	Readable and modifiable with scale-flexible architecture
Speed	Slow performance as it uses Theano or TensorFlow in the backend for processing	Fast performance because of the profiler in backend processing
APIs	Has a high-level API and can run on Theano and CNTK	Has both high-level and low-level APIs
Debugging	Easy to debug	Complex to debug
Dataset	Used for small datasets	Used for massive datasets
Capability	Capable of rapid prototyping and multiple back-end support	Capable of object detection and functionality

Keras vs. Tensorflow vs. PyTorch

Follow this link

https://www.youtube.com/watch?v=DmI58jz2i6w

THANK YOU